

Detecting and Mitigating DDoS Attacks in SDN Environments Using Machine Learning

Ammar Ahmed Khan¹, Muhammad Asim², Muhammad Ahsan Shafique³, Musadique Hussain⁴

Department of Software Engineering, NED University of Engineering and Technology

¹khan4405225@cloud.neduet.edu.pk

²asim4409005@cloud.neduet.edu.pk

³shafique4403293@cloud.neduet.edu.pk

⁴hussain4406846@cloud.neduet.edu.pk

Abstract– In the evolving field of network security, one of the major challenges that continue to be faced is how to defend against DDoS attacks, especially within Software Defined Networking (SDN). This research work presents a holistic approach to detect and mitigate DDoS attacks in SDN environments using Mininet for network simulation, Ryu controller for SDN management and machine learning techniques. Using the CICIDS 2017 dataset, we trained a Random Forest classifier to differentiate between benign and malicious traffic. The idea is for this system to quickly identify anomalies in network traffic and implement mitigation plans that isolate or block traffic sources containing harmful data only without causing much disruption to normal activities by authorized users over the network. Thus, this paper examines how machine learning techniques can be integrated into SDN architecture to improve the resilience of networks against emerging DDoS threats.

I. INTRODUCTION

The world of networking business has gone through a radical shift due to the advent of a new generation of networking models collectively termed Software Networking. Many industry giants or telecoms and networking providers such as Cisco Systems have begun migrating their networking infrastructure to SDN. [1] As opposed to the normal network where the control plane and the data plane are integrated SDN was designed to split the two planes. In SDN the control intelligence and operational taxonomy are held in the ‘control plane’, and the actual enforcing of the rules or decisions is in the ‘data plane’. [2] The applications that are used in the network application layer determine various routing procedures, QoS strategies, and other necessities, provisions, and measures of the network. [2] which is used to forward the information to the control plane by using the north bound interface. Having such policies in place the control plane can derive the routing, the packet forwarding and all the other policies required by the data plane. [3] These are then sent via the southbound interface to the data plane/forwarding plane of the network. The mechanism of sharing this information is usually achieved through the OpenFlow protocol. [4]

However, in conventional networks, control logic and data plane are closely inherent with every device and alters look

very costly and laborious as the network administrator needs to download low-level code into every device. [3] It is very cumbersome and error-prone to manage a network and its configurations manually, especially in large networks. In addition, it is necessary to notice that the management and monitoring of traditional networks occur by the hands of the network administrators. [5] complicating the management of networks and the configuration of policies on the same.

SDN on the other hand, dismantles the control plane from the data plane thus directing the network's controls while making it application programmable. This separation ensures that the network administrators are flexible to set, oversee as well as draw useful analyses of the entire network operations and resources. Rules and policies can be set at the central level and implemented to the extent of the network hence making the process even more flexible and easy [5][6] with limited demands from certain vendors.

The separated control architecture brings new opportunities to introduce innovations and enhancements to our system, as well as uncovers new vulnerabilities in the networking systems. It's a form of threat such as a Distributed Denial of Service (DoS). This attack works by flooding the server with more traffic than the network can handle, thus degrading the performance of the network to the worst and bringing the network to a complete standstill [7]. This is always for a specific device or a service in the traditional networks. However, in SDN, the central control plane is the primary focus, as the set of control plane entities is smaller. An overload or jeopardizing of the control plane can lead to a network failure while at the same time changing the overall nature of the network.

II. RELATED WORK

Traditional or current DDOS attacks are one of the severe threats to the interests of network security, particularly in Software Defined Networks (SDN). Various methods that can be suitable for the identification and countermeasures against DDoS threats have been examined in some studies. This literature review provides a general step-by-step picture of the

advances in this topic together with the used approaches, effectiveness, and challenges.

A. Entropy-Based Approaches

Hong et al. [16] presented one technique for the mitigation of slow HTTP DDoS attacks against targets including servers. Its algorithm operates out of the SDN controller and analyses the TCP connections, finding the attack based upon the count of half-open connections. When the defined traffic influx is attained, the connections are transferred to a mitigation algorithm that distinguishes slow clients from attackers. This approach does prevent connections from known malicious IPs but could add a certain level of delay since it depends on the half-open connection synchronisations.

In a similar capacity, Kalkan et al. [17] used entropy jointly with TCP layer parameters to discern DDoS attacks. Through the probability estimation of typical traffic characteristics such as "Destination IP" and "TTL", they were able to detect rather large entropy during the given attacks. This approach can significantly minimize false positives; however, it has rather heavy computational demands for permanent entropy computation as well as can perform worse in case of highly fluctuating traffic.

In their two-phase detection and prevention architecture given in [15], the authors have utilized Shannon's entropy. The first phase involves using entropy to detect illegal traffic while the second phase involves using machine learning to classify the flows. Although this method improves the detection rate, it becomes resource demanding and thus can slow down the continuous traffic analysis in real-time.

B. Machine Learning-Based Approaches

In [10] Li et al. introduced a machine learning framework for SDN security. The authors gathered traffic data and then preprocessed extracting features from the collected data including the IP addresses as well as protocol types using entropy to get the features distribution. From the analysis of this data, a nonlinear SVM classifier approached higher accuracy as compared to the other ML methods such as DT, NB, KNN, and RF. However, this approach mainly deals with detection and is not very effective regarding real-time mitigation.

Dennis et al. [11] proposed the DDoS detection method based on flow analysis by applying the RF classifier method with weighted voting. Their approach included flow statistics sampling at the time interval of 10 seconds and categorizing traffic based on such parameters as the number of packets and flow's duration. When an attack was picked, the system cut off the hostile traffic. It is an effective approach of detection followed by mitigation though it uses flow features

which when sometimes may cause the model to fail to distinguish between normal and malicious traffic.

Ye et al. [12] described the [insert type of ML algorithm] based detection scheme which gathers traffic data at regular time intervals and calculates accumulated features characteristic to DDoS attacks. This method is very useful in accurately detecting cases, but the abovementioned solutions are not very systematic, thus the utilization of this method is somewhat constrained.

Oo et al. [13] extended the above SVM technique with an Advanced SVM (ASVM) algorithm to increase the detection accuracy with extra accumulated traffic characteristics. However, similar to other methods based on SVM, the approach successfully increases the identification results but remains insufficient in the aspect of managing them.

Rahman et al. [14] proposed an SDN framework with incorporates several ML methods to prevent and identify DDoS attacks. The authors built up a dataset consisting of 24 features of the packet layers and they selected J48 as the optimal classifier. Their framework had a mitigation script that involved denying the attacker ports for some time. It rivals the previous method as far as detection interconnectivity is concerned, but it may cause high processing and memory overhead because of the reliance on packets.

Alshamrani et al. [9] proposed offline and online DDoS detection with the help of a secure system using SVM. They used the NSL-KDD datatype to train their model and as for implementation, they used it as a REST API at the SDN controller. In the case of an attack, it was decided that the malicious traffic would be diverted towards a honeypot. Although this increases the chances of detection and gives a detailed analysis of the attack, this approach introduces a new problem of time delay in handling real-time traffic.

C. Deep Learning-Based Approaches

Rasool et al. [8] looked into the applicability of deep learning to classify traffic in the SDN architecture. Some of the features used were link utilization rates, number of packets per flow, and packet size; They conducted a Link Flooding attack and tested an Artificial Neural Network Classifier on the differentiation between safe and risky traffic flow. Although deep learning adds precision to the classification, its delays take it out of the real-time applications category, it requires a lot of computing power.

TABLE I
SUMMARY OF THE LITERATURE REVIEW

Authors	Computational Overhead	Latency Issues	Mitigation Strategies	Scalability Concerns
Hong et al. [16]	Low - Simple monitoring of TCP connections	Moderate - Threshold-based detection may delay response	Effective for slow HTTP attacks, but relies on thresholds	Limited to handling specific attack types
Kalkan et al. [17]	High - Continuous entropy calculations	Low - Entropy calculation is quick but resource-intensive	Lacks real-time mitigation; focuses on detection	Struggles with highly dynamic traffic patterns
Da Silva et al. [15]	High - Two-phase process with entropy and ML	High - Entropy and ML analysis can delay response	Accurate detection, but mitigation may be delayed	Computationally intensive, not easily scalable
Li et al. [10]	Moderate - Entropy and SVM processing	Moderate - SVM training and classification on time	Effective detection; limited real-time mitigation capability	Focuses on detection, less on mitigation
Dennis et al. [11]	Moderate - Regular flow statistics collection	Low - Flow-based analysis reduces latency	Combines detection with immediate traffic-dropping	Depends on flow features; may miss some attacks
Ye et al. [12]	Moderate - Aggregated feature extraction	Low - Periodic data collection minimises latency	Accurate detection but lacks detailed mitigation strategies	Can struggle with real-time scalability
Oo et al. [13]	High - Advanced SVM with additional features	Low - Enhanced detection but no real-time focus	Enhanced detection capabilities, but mitigation is weak	Better detection but lacks in mitigation scope
Rahman et al. [14]	High - Packet-based feature	High - Packet-level analysis	Effective integration of detection	High processing and memory

	extraction	induces delays	and mitigation but resource-heavy	overhead
Alshamrani et al. [9]	Moderate - SVM model processing	Moderate - Redirection to honeypot introduces delays	Detection with detailed analysis via honeypot; may delay actual handling	May face delays in real-time processing
Rasool et al. [8]	Very High - Deep learning model training and inference	High - Processing delays due to deep learning	High accuracy in classification but not practical for real-time	Demands significant computational resources

III. SYSTEM ARCHITECTURE

Our system can recognize DDoS attacks and SDN can mitigate them. In our suggested architecture, several modules work hand in hand to enable network traffic monitoring, anomaly detection as well as the proper mitigation measures. It not only detects and mitigates DDoS attacks but also ensures that valid traffic is not disturbed.

Our proposed SDN architecture is divided into three layers:

A. Architectural Layers

- 1) *Application Layer*: This layer consists of network applications that offer special services, particularly in addressing security, traffic flow and control, network surveillance and so on. In this work, the DDoS detection and mitigation module is at the application layer for this project. Then it interacts with the SDN controller to obtain the network statistics and to apply the security policies.
- 2) *Control Layer*: The Control Layer is controlled by the Ryu controller, which is the central unit in the SDN network. It manages the data communication through interactions with devices embedded in the infrastructure layer by OpenFlow protocol. Ryu controller receives flow statistics from switches, processes requests coming from the application layer and installs the flow rules for traffic management.

- 3) *Infrastructure Layer*: This layer is composed of physical components such as routers and switches that await instructions from the controller on where to direct packets. In our implementation, Mininet emulates this layer by creating virtual network topologies which can be controlled programmatically.

B. Modular Components

The system architecture can be divided into several modular sub-systems. All the modules play particular roles in detecting and mitigating DDoS attacks.

- 1) *Flow Collector Module*: The Flow Collector module is to acquire flow statistics data in the network switches. This process involves using the OpenFlow protocol to send flow-stats requests to the intended switches with a view to getting the desired information. Every switch returns the flow table entries along with packet count, byte counts as well as the flow duration. These entries are very important for studying the traffic in the network. Flow Collector offers these responses to filter actual flow data such as flow-id and its related counters eliminating the other information is not required. This collected data is used in the subsequent analysis and detection processes.
- 2) *Feature Extender Module*: The Flow Collector module is expected to collect flow statistics from the network switches. This can be achieved by initiating flow-stats requests using the OpenFlow protocol on each of the switches in the network. For each switch, OpenFlow sends backflow entries that include packet count, byte count, and flow duration among others. These entries are essential in determining the nature of the network traffic. The responses are then processed by the Flow Collector to obtain the fundamental flow details such as the flow identification number (flow-id) and other counters while dismissing unnecessary information. This collected data is used as input for further analysis and detection activities.
- 3) *Anomaly Detection Module*: This is where the security mechanism of the system is challenged and forced to act in providing necessary measures. The module has used a Random Forest as one of the machine learning methods in the classification of the network flows into normal and abnormal. Random Forest was chosen as the model of interest on account of its performance in high-dimensional datasets and the datasets formed by Feature Extender would be high-dimensional. It

monitors flows over the network in subsequently real-time to analyze and label the incoming ones as potential cases of DDoS attacks. It's perhaps the classification that happens in no time which helps in the formulation of quick responses to emergent risks.

- 4) *Anomaly Mitigation Module*: Once an anomalous flow is identified, the Anomaly Mitigation module then intervenes to stop the ongoing DDoS attack that is likely to cause havoc on the network. This module dynamically communicates with the Ryu controller to install flow entries that prevent traffic originating from the identified malicious sources. In the current implementation, the mitigation strategy is to blacklist the Ethernet address about the malicious flow. This approach is efficient in ending the attack at the source level without disrupting the other normal traffic in the networks.

The modular architecture allows the system to efficiently detect abnormal traffic and take necessary actions without disrupting the legitimate traffic. The integration of machine learning makes the system more efficient, robust and adaptable to evolving DDoS threats.

IV. DETECTION AND MITIGATION METHODOLOGY

This part will discuss the working procedure of our system. This should provide information on how the system identifies the DDoS attacks through machine learning and how our system is built to prevent them from the network.

A. Packet Collection and Feature Extraction

- 1) *Flow Collection*: It starts with the Flow Collector module, which is responsible for collecting flow statistics from all switches in the network at regular intervals. Some of these are packet count, byte count, and flow duration which are crucial in the analysis of traffic. In the OpenFlow protocol, the Flow Collector verifies that all the necessary and complete information about each and every network flow is captured perfectly.
- 2) *Feature Extraction and Enhancement*: The feature Extender module operates on the collected flow data although the flow data is collected by another feature. This requires obtaining source and destination IP addresses and MAC addresses, ports, and protocol types of each of the flows. Also, derived features such as per packet transmission rate, the flow duration and

so on are derived. These other features go a long way in helping in the differentiation of normal traffic as from the malicious ones.

B. Detection and classification:

- 1) *Machine Learning Model:* Random forest classifier is the core of the detection mechanism in the given steps. It is trained from a dataset that belongs to normal and also to the flow of traffic containing some sort of danger. Therefore random forest algorithm is fine tuned for this purpose as it can handle large number of features and does not build up a model specifically for the given data. In this case during the training process it is trained on how to distinguish a DDoS attack from normal traffic intensity. Therefore, in real time, to the traffic analysis these patterns are applied and the classifier defines each flow as normal or anomalous.
- 2) *Real-time Classification:* As a result of the improved incoming flows, during the operation phase, they are input into the trained Random Forest model. The obtained values are then compared to the criteria learned by the model and each flow is assigned to one of the categories. The other flows are tagged as suspicious if they resemble the types of flows that are typical in previous DDoS attacks. Due to this real-time classification, any form of threat is ferreted out and addressed within the shortest time possible.

C. Mitigation Strategy

- 1) *Alert and Analysis:* In case of identification of any doubtful flow, the Anomaly Detection module informs the Anomaly Mitigation module. It includes a comprehensive report about the observed abnormality including its source, destination addresses as well as its type. This information is processed by the mitigation module to make the right decision to be taken.
- 2) *Blocking Malicious Traffic:* The primary mitigation strategy is to avoid traffic from the source of the identified DDoS attack. In our system, this is done by commanding the Ryu controller to install flow rules that cause packet from the said malicious source to be dropped. The controller is able to impose these rules on all the switches in the network hence blocking the attack at its source. This method helps in making sure that the bad traffic is stopped without affecting the normal, good traffic.

- 3) *Dynamic Rule Enforcement:* The mitigation module constantly changes the flow rules according to the detected threats because these threats are constantly changing as well. If the attack continues or morphs into a different type of attack, the system can alter mitigation rules to accommodate the new form of attack. This dynamic response capability is important, especially in combating complex DDoS attacks which may try to bypass static security measures.
- 4) *Monitoring and Adaptation:* Once the mitigation is done, the system keeps on scrutinizing the network traffic. It recalculates the flow data to determine the efficacy of the mitigation measures and to identify new or continuing risks. This continuous monitoring ensures that the network remains secure and that normal operations are maintained with minimal disruption.

V. EVALUATION AND RESULTS

The evaluation phase of the presented DDoS detection and mitigation system has deliberately followed a methodical and systematic approach in order to guarantee that the used approach is both sound and effective. Our target was to assess our system under conditions that would realistically represent real-life possible situations with the help of the CICIDS 2017 dataset, which has a variety of both, normal and cyber attacks. Below is the detailed breakdown of the results outlining the degree of success by our system throughout the tests and different conditions.

A. Experimental Setup and Environment

Due to the demand for a highly realistic network, we emulated the network in Mininet from top to bottom using multiple hosts and switches. This network was controlled by the Ryu SDN controller, which was used for the detection of the network event and control of the network events. This configuration enabled us to firstly recreate the circumstances that lead to DDoS attacks and secondly, enabled to monitor the system response in real time.

For the data set, CICIDS 2017 dataset was used in the evaluation of this paper. This dataset contains various flows from the network traffic to and from different hosts, and the specific features of each flow distinguishes between normal traffic and traffic from a distributed denial of service attack. To improve the validity of our detection model, we divided this data into two sets; the training data set for training the model and the testing data set for testing the model.

B. Performance Metrics

The following performance measures were chosen as the most important ones to assess the effectiveness of the developed system in protecting from and identifying DDoS attacks.

- 1) *Detection Accuracy*: The classifier we integrated into the system of our study was the Random Forest classifier which was trained on 70% of data in the CICIDS 2017 dataset and was then tested on the remaining 30% of the dataset. The proposed model achieved a detection accuracy of about 98.7%. Accuracy is expressed by the ratio of correctly classified instances including both the benign and the malicious ones to the total instances. This high accuracy proves the effectiveness of the chosen model along with its ability to separate normal traffic from the attack traffic.
- 2) *Precision*: Our system achieved a precision rate of 97.9%, meaning that out of all traffic flows identified as malicious, 97.9% were DDoS attacks. From this score in high precision, it is thus evident that this system incurs a very minimal number of false alarms. This is important because, with so few false positive rates, it can guarantee that actual traffic is not hampered in any way.
- 3) *Recall*: Our system had a recall rate of 98.4%, which indicates that it would be able to detect most of the DDoS out of the total number of malicious flows existing in the dataset adopted for training. High Recall means that in the learning context, the majority of DDoS attacks would be noticed by our system, and there would be little leakage.
- 4) *F1-Score*: An F1-score of 98.1% was obtained by calculating the harmonic mean between precision and recall. The definition of this metric suggests that we do not miss opportunities and do not introduce false positives in the process, which is embedded in the design; this is also very handy when both precision and recall are imperative.

C. Detection Time

The ability of our system to identify DDoS attacks is important so that it can take necessary steps before the attacks escalate. In all of the performed experiments the system was able to identify anomalous traffic within the range of 15 to 20 seconds from the beginning of an attack. This rapid detection is made possible by the ability of the system to process data and the quick classification of the Random Forest algorithm over

the network flows. Awareness is critical in mitigating damages of DDoS attacks since the attacker and the intended target may take action as soon as the attack is detected.

D. Mitigation Effectiveness

When a DDoS attack is identified on a client's network, our system immediately activates the Anomaly Mitigation module. This module is charged with the role of identifying and preventing the traffic with the contaminated data. The results of the tests suggested that the system was capable of applying the mitigation measures response within an average of 2 minutes of the detection of an issue. The mitigation process involved:

- 1) *Source Blocking*: The system then isolates the Ethernet address belonging to the source of the malicious traffic and hence prevents the subsequent attack traffic that originated from that source but allows legitimate traffic.
- 2) *Minimal Disruption*: Our mitigation approach was such that normal network traffic was not disrupted, hence, legitimate use of the network remained functional even as the mitigation was undergoing.

The comprehensive evaluation of our DDoS detection and mitigation system demonstrates its effectiveness and reliability in identifying and neutralizing DDoS attacks in a Software-Defined Networking context.

VI. CONCLUSION AND FUTURE WORK

This research presented and assessed a powerful DDoS detection system that was integrated for the context of the SDN environment. There is a dataset known as CICIDS 2017 with which we trained a Random Forest classifier as it has been found perfect for swift detection. This made the job easy for the network system to tell the normal traffics from the problematic ones. Also, the response offered by the mitigation process of the system was considerable as it ensured its provision of quick response while causing disruptions of legitimate network operational activities to a minimal level. As we demonstrated during our evaluation, this concept is not only practical but is larger to true precise community safety applications.

For the future work, we will design the improvement of the system's capabilities by employing the more advanced features of machine learning and expanding the variety of detected attacks. As for us, the same goals include further optimization of measures for addressing complex attack scenarios and explore the application of deep learning models

to achieve higher accuracy with regard to positive and negative findings. Furthermore, the studies with more complicated or extensive and large-scale networks will be required to support or expand the range of its applicability and efficiency under the various realistic conditions.

REFERENCES

- [1] M. Dabbagh, B. Hamdaoui, M. Guizani and A. Rayes, "Software-defined networking security: pros and cons," in *IEEE Communications Magazine*, vol. 53, no. 6, pp. 73-79, June 2015, doi: 10.1109/MCOM.2015.7120048
- [2] E. Kaljic, A. Maric, P. Njemcevic and M. Hadzialic, "A Survey on Data Plane Flexibility and Programmability in Software-Defined Networking," in *IEEE Access*, vol. 7, pp. 47804-47840, 2019, doi: 10.1109/ACCESS.2019.2910140.
- [3] M. Paliwal, D. Shrimankar and O. Tembhurne, "Controllers in SDN: A Review Report," in *IEEE Access*, vol. 6, pp. 36256-36270, 2018, doi: 10.1109/ACCESS.2018.2846236.
- [4] N. McKeown et al., "OpenFlow: Enabling innovation in campus networks", *ACM SIGCOMM Comput. Commun. Rev.*, vol. 38, no. 2, pp. 69-74, Apr. 2008.
- [5] M. Awais, M. Asif, M. B. Ahmad, T. Mahmood and S. Munir, "Comparative Analysis of Traditional and Software Defined Networks," *2021 Mohammad Ali Jinnah University International Conference on Computing (MAJICC)*, Karachi, Pakistan, 2021, pp. 1-6, doi: 10.1109/MAJICC53071.2021.9526236.
- [6] T. Wang, "Benefits and the Security Risk of Software-defined Networking," www.isaca.org, Jul. 01, 2016. <https://www.isaca.org/resources/isaca-journal/issues/2016/volume-4/benefits-and-the-security-risk-of-software-defined-networking>.
- [7] N. Goksel and M. Demirci, "DoS Attack Detection using Packet Statistics in SDN," 2019 International Symposium on Networks, Computers and Communications (ISNCC), Istanbul, Turkey, 2019, pp. 1-6, doi: 10.1109/ISNCC.2019.8909114.
- [8] R. U. Rasool, U. Ashraf, K. Ahmed, H. Wang, W. Rafique and Z. Anwar, "Cyberpulse: A Machine Learning Based Link Flooding Attack Mitigation System for Software Defined Networks," in *IEEE Access*, vol. 7, pp. 34885-34899, 2019, doi: 10.1109/ACCESS.2019.290423
- [9] A. Alshamrani, A. Chowdhary, S. Pisharody, D. Lu, and D. Huang, "A Defense System for Defeating DDoS Attacks in SDN based Networks," *Proceedings of the 15th ACM International Symposium on Mobility Management and Wireless Access - MobiWac '17*, 2017, doi: <https://doi.org/10.1145/3132062.3132074>.
- [10] Li, Dong & Yu, Chang & Zhou, Qizhao & Yu, Junqing. (2018). Using SVM to Detect DDoS Attack in SDN Network. IOP Conference Series: Materials Science and Engineering. 466. 012003. 10.1088/1757-899X/466/1/012003.
- [11] Dennis, J. Rukshan, X. Li. "Machine-Learning and Statistical Methods for DDoS Attack Detection and Defense System in Software Defined Networks", 2018.
- [12] J. Ye, X. Cheng, J. Zhu, L. Feng and L. Song, "A DDoS Attack Detection Method Based on SVM in Software Defined Network", *Security and Communication Networks*, vol. 2018, pp. 1-8, 2018.
- [13] M. Myint Oo, S. Kamolphiwong, T. Kamolphiwong and S. Vasupongayya, "Advanced Support Vector Machine- (ASVM-) Based Detection for Distributed Denial of Service (DDoS) Attack on Software Defined Networking (SDN)", *Journal of Computer Networks and Communications*, vol. 2019, pp. 1-12, 2019
- [14] O. Rahman, M. A. G. Quraishi and C. Lung, "DDoS Attacks Detection and Mitigation in SDN Using Machine Learning," 2019 IEEE World Congress on Services (SERVICES), Milan, Italy, 2019, pp. 184-189.
- [15] da Silva, Anderson Santos, Juliano Araujo Wickboldt, Lisandro Zambenedetti Granville, and Alberto Schaeffer-Filho. "ATLANTIC: A framework for anomaly traffic detection, classification, and mitigation in SDN." In *NOMS 2016-2016 IEEE/IFIP Network Operations and Management Symposium*, pp. 27-35. IEEE, 2016.
- [16] Hong, Kiwon, Youngjun Kim, Hyungoo Choi, and Jinwoo Park. "SDNassisted slow HTTP DDoS attack defense method." *IEEE Communications Letters* 22, no. 4 (2017) 688-691.
- [17] K. Kalkan, L. Altay, G. Gur, and F. Alagoz, "JESS: Joint Entropy-Based DDoS Defense Scheme in SDN," *IEEE Journal on Selected Areas in Communications*, vol. 36, no. 10, pp. 2358–2372, Oct. 2018, doi: <https://doi.org/10.1109/jsac.2018.2869997>.